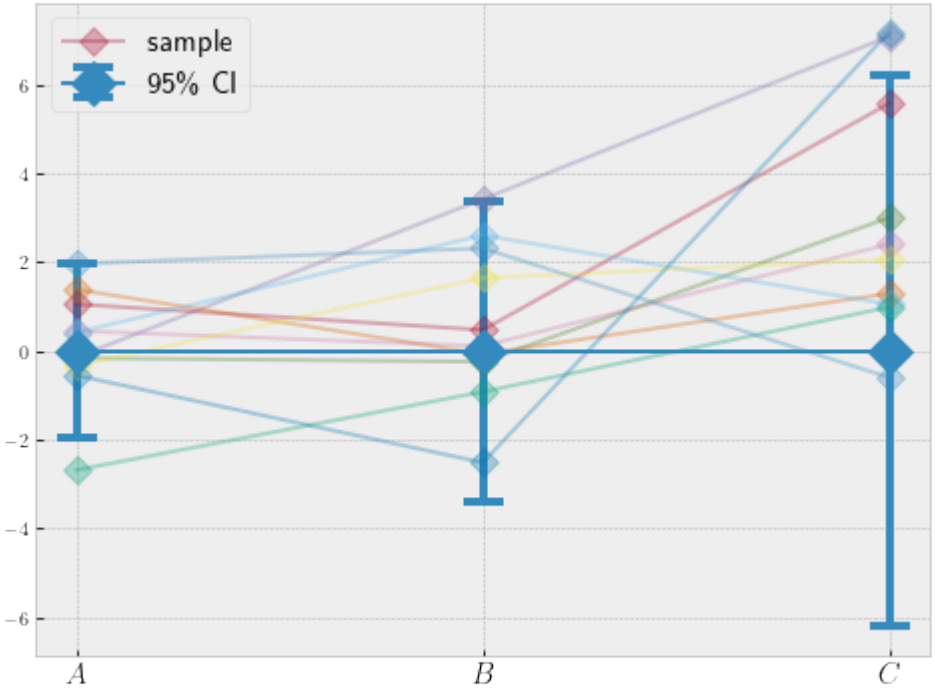


Figure 2.5 Parallel coordinates plot visualizing a mean-normalized multivariate Gaussian from the housing price example. The error bars indicate 95% credible intervals of corresponding normal distributions, while the faded lines show samples drawn from the multivariate Gaussian.

Note the dark diamonds and corresponding error bars.

- The diamonds represent the mean vector of our Gaussian, which is simply the zero vector.
- The error bars denote the 95% credible intervals, or CIs, of the three individual variables. Going from A to B to C, we observe larger and larger CIs, corresponding to the increasing values of the respective variances.

SIDEBAR **Credible intervals**

A (1 -) credible interval (CI) of a normal distribution of a random variable x is a range in which the probability that x falls inside this range is exactly (1 -).

Statisticians typically use 95% CIs for normal distributions. There is nothing really special about the number 95% here, expect for the fact that 95% is the threshold that many statistical procedures use to determine whether something is meaningful or not. For example, a t-test typically uses a confidence level of 1 - = 0.95, corresponding to the fact that a p-value less than = 0.05 indicates significant results. A convenient fact about normal distributions is that ± 1.96 is a 95% CI (some actually even use ± 2), where and are the mean and standard deviation of the variable x, which is an easy-to-compute quantity.

Figure 2.5 represents our *prior belief* about the normalized prices of the three houses. Starting

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